MIL-P-60019A (MU)
15 January 1969
SUPERSEDING
MIL-P-60019 (MU)
31 January 1964

MILITARY SPECIFICATION

PROPELLANT, M9
FOR USE WITH CHARGE, PROPELLING M36A1
FOR 4.2 INCH MORTAR

1. SCOPE

1.1 This specification covers one type of flake propellant designated as M9 for use in charge, propelling M36A1 for the 4.2 inch mortar, M30.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	-	Ammunition and Special Weapons; General Specification for
MIL-I-45208	-	Inspection Requirements; Supply and Maintenance

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-414	-	Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-652	-	Propellants, Solid, for Cannons; Requirements and Packing
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1376

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DRAWINGS

ARMY

20-4-616	Marking Diagram and Sealing of Steel Packing Boxes for Shipment of Propellants
76-4-46	Box, Packing, with Metal Liner, M24, for Smokeless Powder, Assembly and Details
76-4-53	Box, Steel, M2, for Smokeless Powder Assembly
76-4-56	Box, Packing, Metal-Wood, M17 for Smokeless Propellant
7549033	Container, Metal, Universal, M25 for Propellants and Explosives, Assembly and Details
8858577	Marking Diagram and Sealing of Container, Metal, Universal, M25, for Shipment of Propellant
8858848	Marking Diagram and Sealing of Metal Lined Wooden Packing Boxes for Shipment of Propellant
8863614	Bag Loading Assembly
8863617	Charge, Propelling, M36A1 Assembly

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 171-179

The Code of Federal Regulations is available from the Superintendent of Documents, Government Printing Office, Washington, D.C., 20402. Orders for the above publication should cite: 49 CFR 171-179 (latest revision).

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3. REQUIREMENTS

3.1 Material.-The propellant shall be M9 flake propellant with an advisory diameter of 0.059 inches, a maximum thickness of 0.010 inches, and shall conform to the requirements of Standard MIL-STD-652 (see 4.2.5).

3.2 Proving Ground.-The propellant, when fired in the weapon and with the weight of projectile for which it was manufactured, shall comply with the following requirements when tested as specified in 4.3.1.

3.2.1 Velocity.-The weight of propellant shall be established by charge assessment which will impart a muzzle velocity of 355 feet per second (fps) (108.2 meters per second) at 70 degrees Fahrenheit (°F).

3.2.2 Uniformity of velocity.-The standard deviation in velocity during the Velocity Uniformity series shall not exceed 5.0 fps factored or 6.8 unfactored (see 6.5).

3.2.3 Pressure.-No individual pressure experienced during the Velocity Uniformity test series shall exceed 4300 psi or be less than 2900 psi.

3.3 Graphite (Glaze).-The graphite content shall not exceed 0.20 percent when determined as specified in 4.3.4.

3.4 Workmanship.-The best commercial practices shall be used in the formulation of propellant furnished under this specification and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. No reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification unless otherwise specified.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein.

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4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.

b. Number of batches of product inspected (shall be made available upon request by the contracting officer).

c. Results obtained, by defect code, for all inspections performed (shall be made available upon request by the contracting officer).

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents (shall be made available upon request by the contracting officer).

f. Number of items in the lot.

g. Date submitted.

h. Eight copies of propellant description sheet AMC Form 1047 (00 Form 1204) containing the applicable data (see 6.6).

4.1.2 The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection provisions

4.2.1 Lot formation.-The term "lot" as used throughout this specification, refers to an inspection lot, which is defined as an essentially homogeneous quantity of propellant from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. A lot shall consist of one or more sub-lots of M9 propellant produced by one manufacturer, in accordance with the same specification, and same specification revision under one continuous set of operating conditions. Each lot shall consist of that quantity of M9 propellant which has been subjected to the same unit chemical or physical process intended to make the final product homogeneous.

4.2.2 Examination.-Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans and procedures for major and minor defects shall be in accordance with Standard MIL-STD-105 except that continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed, using an AQL of 0.65 percent for

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each minor defect and an AQL of 0.40 for each major defect, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with paragraph 4.2.4.

4.2.2.1 Containers, prior to closing (see drawing 76-4-46, 76-4-53, and 76-4-56).

Categories	Defects	Method of Inspection	Code No. (See 6.4)
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Crack or hole in metal liner (as applicable)	Visual	01001
102.	Foreign matter or rust on interior	Visual	01002
Minor:	None defined.		

4.2.2.2 Sealed container (see drawing 76-4-46, 76-4-53 and 76-4-56).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Gasket missing or broken	Visual	02001
102.	Wooden box broken or split (as applicable)	Visual	02002
103.	Cover improperly closed	Visual-Manual	02003
Minor:	AQL 1.50 percent		
201.	Hardware improperly engaged	Visual	02004
202.	Markings misleading or unidentifiable	Visual	02005
203.	Bare areas in exterior coating the sum of which is in excess of 1/4 square inch	Visual	02006
204.	Car seal missing, unsealed or improperly positioned	Visual-Manual	02007

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4.2.3 Testing

4.2.3.1 Proving Ground (see 3.2).-Each lot of propellant shall be sampled for ballistic acceptance in accordance with MIL-STD-652 with one exception: ten samples shall be selected for ballistic testing.

4.2.3.1.1 Uniformity of velocity (see 3.2.2), Major defect, Code No. 04001.-If the product of the factor 0.73 times the velocity standard deviation (see 6.5) of the test rounds fired at 70°F exceeds the applicable requirement, the lot shall be rejected.

4.2.3.1.2 Pressure (see 3.2.3), Critical defect, Code No. 03001.-A lot of propellant shall be rejected if any individual test round(s) at the five increment level fails to meet the applicable pressure requirement.

4.2.3.2 Retest.-If for any reason the proving ground considers that test conditions have detrimentally affected the test results, the test shall be declared invalid and a new test performed on additional charges as required.

4.2.3.2.1 If the difference between the final recommended charge weight and the charge weight employed in the 5 increment velocity uniformity series exceeds 10 grains, the 10 round uniformity test shall be repeated with the recommended charge weight.

4.2.4 Inspection equipment.-The contractor shall have available, and utilize correctly, all necessary standard gages, test and measuring and equipment required for the performance of the inspections specified herein.

4.2.5 Flakes (see 3.1), Major defect, Code No. 05001.-If the average flake thickness exeeeds 0.010 inches, the lot shall be rejected.

4.3 Test methods and procedures

4.3.1 Acceptance tests.-These tests shall be conducted at a Government Proving Ground in accordance with the Aberdeen Proving Ground Acceptance Test Procedure, P-VAR, Supplement 4.2M-2.

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4.3.2 Charge assessment.-Sufficient groups of rounds, each loaded with a different weight of propellant, shall be conditioned at 70°F. After conditioning, the charges shall be fired to establish the charge velocity and charge pressure curves. From the data obtained, the charge weight shall be established which will yield the prescribed service velocity and corresponding expected peak pressure. After ten (10) lots of a given production series have been fired to provide a good knowledge of the propellant characteristics, this phase may be discontinued and the charge weight determined by the constant-slope method using, in addition, rounds specified in 4.3.3.1.

4.3.3 Uniformity series

4.3.3.1 Five increments.-Ten (10) rounds shall be loaded to the charge weight established in 4.3.2, conditioned at 70°F, and fired at the five increment level with each consecutive lot of propellant produced. Ten (10) standard reference rounds shall be similarly loaded, but with reference propellant, conditioned, and fired. No more than five minutes shall elapse from the time the round is removed from the conditioning chamber until it is fired. The temperature of the day at the time of firing shall be reported. The velocity mean and standard deviation shall be calculated after reducing results to standard conditions. All firing data, including individual gauge pressure readings, shall be reported to Picatinny Arsenal for evaluation (see 6.3).

4.3.3.2 Seven increments.-A series of ten (10) rounds shall be fired at the seven (7) increment level in accordance with the procedure specified in 4.3.3, for information only. Both reference and test rounds shall be loaded with the same lot of standard M8 propellant at seven increments.

4.3.4 Graphite (glaze) determination.-The graphite content shall be determined in accordance with Method 308.1 of MIL-STD-286.

4.3.5 Flakes.-One hundred flakes shall be examined under a microscope for form, length, and thickness. The 100 flake sample shall be taken from the vessel containing the entire ballistic sample. (see 4.2.5).

5. PREPARATION FOR DELIVERY

5.1 Packing.-The packing shall be in accordance with MIL-STD-652.

5.2 Marking.-The marking shall be in accordance with MIL-STD-652.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the following:

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a. Title, number and date of this specification.
b. Packing required (see 5.1).
c. Process.-Details of the manufacturing process and the equipment used by the contractor will be submitted to the procuring activity in writing prior to commencement of manufacture.

6.2 Mean and standard deviation.-The mean and standard deviation shall be calculated in a standard statistical technique equivelant to that shown in Standard MIL-STD-414, Section B, Example B-1. (n-1) shall be used as a divisor in calculating the standard deviation.

6.3 Submission of test data.-In addition to the normal distribution of test records, when the propellant is procured by the Department of the Army, one copy of all ballistic test data shall be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND2, and one copy to ATTN: SMUPA-DE2, Dover, New Jersey, 07801.

6.4 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.5 Factor 0.73.-The factor 0.73 makes allowance for the probability that a standard deviation from a ten-round sample exceeds the true standard deviation by chance alone.

6.6 Submission of AMC Form 1047.-Information contained in propellant description sheet, AMC Form 1047, need not be resubmitted in the form of a certificate provided that the provisions regarding the certifying of responsible agents are adhered to (see 4.1.1).

6.7 Tolerance.-A tolerance of plus or minus 2 degrees F applies to all 70 degrees F requirements.

6.8 Surveillance samples.-Samples shall be forwarded to Picatinny Arsenal, Dover, New Jersey, ATTN: SMUPA-VG, for the propellant surveillance test program in accordance with Standard MIL-STD-652.

6.9 Proving Ground test summary

6.9.1 Charge assessment.-Propellant to be assessed to Muzzle Velocity of 355 fps at 70°F. Number of rounds required shall be determined by Proving Ground.

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6.9.2 Uniformity Series

Temperature	Charge Increment	Number of Samples	Std. Dev. of MV (Factored)	Individual Pressure
70 degrees	5	10	5.0 fps (max.)	2900 psi-4300 psi
70 degrees	7	10	INFO	INFO

Custodian:
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Preparing Activity:
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